

Course: Full Stack Web Development

Course Code: 23CM4121

Task No.: 1.2

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Title

Design Webpage Using HTML5 & CSS3

Aim

To design and develop a styled webpage using HTML5 and CSS3 by applying semantic HTML elements, CSS properties, layouts, colors, fonts, borders, shadows, hover effects, and responsive card-based design.

OBJECTIVE

The objectives of this experiment are:

- To understand the basic concepts of HTML5 and CSS3.
 - To use semantic HTML5 elements for structuring a webpage.
 - To apply CSS properties for styling webpage elements.
 - To understand colors, fonts, margins, padding, borders, and shadows.
 - To create a flexible card-based layout using CSS Flexbox.
 - To implement hover effects on links and buttons.
 - To use gradients for creating attractive backgrounds.
 - To understand responsive layouts using flex-wrap.
 - To design a simple and user-friendly webpage.
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THEORY

HTML5 (HyperText Markup Language 5) is used to structure the content of a webpage. It provides semantic elements such as <header>, <nav>, <main>, <section>, and <footer> that clearly define different parts of a webpage.

CSS3 (Cascading Style Sheets 3) is used to control the appearance and layout of HTML elements. CSS3 provides properties for applying colors, fonts, spacing, borders, shadows, gradients, flexible layouts, and interactive effects.

In this experiment, a webpage titled "My Styled Webpage" is developed. The webpage contains a header, navigation bar, three content cards for HTML5, CSS3, and JavaScript, and a footer.

The webpage uses CSS Flexbox to arrange the cards. The flex-wrap property allows the cards to move to the next line when sufficient space is not available, making the layout more responsive.

Important CSS properties used in the experiment include:

- background – Applies background styling.
- linear-gradient() – Creates a gradient background.
- color – Changes the text color.
- font-family – Specifies the font style.
- text-align – Aligns text.
- padding – Adds space inside an element.
- margin – Adds space outside an element.
- display: flex – Creates a flexible layout.
- flex-wrap – Allows elements to move to the next line.
- gap – Creates space between flex items.
- border – Adds a border around an element.
- border-radius – Creates rounded corners.
- box-shadow – Adds shadow around elements.
- text-shadow – Adds shadow to text.
- :hover – Applies styling when the pointer is placed over an element.

ALGORITHM / PROCEDURE

1. Create a new HTML file.
2. Define the basic HTML5 document structure.
3. Create a header section containing the webpage title.
4. Create a navigation bar with Home, About, Courses, and Contact links.
5. Create the main content section.
6. Create three cards for HTML5, CSS3, and JavaScript.
7. Add headings, paragraphs, and buttons inside each card.
8. Create a footer containing the copyright information.

9. Create a CSS file for styling the webpage.
10. Set the body margin, font, and gradient background.
11. Style the header and navigation bar.
12. Use Flexbox to arrange the cards horizontally.
13. Apply borders, rounded corners, padding, margins, and shadows to the cards.
14. Apply hover effects to navigation links and buttons.
15. Use flex-wrap to make the card layout responsive.
16. Style the footer section.
17. Open the HTML file in a web browser.
18. Verify the webpage layout, colors, cards, buttons, and hover effects.
19. Perform multiple test cases.
20. Compare the expected output with the actual output.

Code

HTML CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Styled Webpage</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <!-- Header -->
  <header>
    <h1>My Styled Webpage</h1>
  </header>

  <!-- Navigation -->
  <nav>
    <a href="#">Home</a>
    <a href="#">About</a>
    <a href="#">Courses</a>
    <a href="#">Contact</a>
  </nav>

  <!-- Content Cards -->
  <section class="container">

    <div class="card">
      <h2>HTML5</h2>
      <p>HTML5 provides semantic elements to structure web pages effectively.</p>
      <button>Read More</button>
    </div>

    <div class="card">
      <h2>CSS3</h2>
      <p>CSS3 is used to style web pages with colors, layouts, animations, and effects.</p>
      <button>Read More</button>
    </div>

    <div class="card">
      <h2>JavaScript</h2>
      <p>JavaScript makes web pages interactive by adding dynamic functionality.</p>
      <button>Read More</button>
    </div>

  </section>

  <!-- Footer -->
  <footer>
    <p>©Copy, 2026 My Styled Webpage. All Rights Reserved.</p>
  </footer>

</body>
</html>
```

CSS Code:

```
/* Body */
body{
  margin:0;
  font-family:Arial, sans-serif;
  background:linear-gradient(to right,#74c8d5,#ACB6E5);
}

/* Header */
header{
  background-color:#2c3e50;
  color:white;
  text-align:center;
  padding:20px;
  text-shadow:2px 2px 5px black;
}

/* Navigation */
nav{
  background-color:#34495e;
  padding:15px;
  text-align:center;
}

nav a{
  color:white;
  text-decoration:none;
  margin:20px;
  font-size:18px;
  font-weight:bold;
}

nav a:hover{
  color:yellow;
}

/* Cards Container */
.container{
  display:flex;
  justify-content:center;
  gap:20px;
  flex-wrap:wrap;
  margin:30px;
}

/* Individual Card */
.card{
  width:250px;
  background:white;
  border:2px solid gray;
  border-radius:15px;
  padding:20px;
  margin:10px;
  box-shadow:5px 5px 10px gray;
  text-align:center;
}

.card h2{
  font-family:Georgia, serif;
  font-size:24px;
}
```

```
.card p{
  font-size:16px;
}

/* Button */
button{
  background-color:blue;
  color:white;
  border:none;
  padding:10px 20px;
  border-radius:8px;
  cursor:pointer;
}

button:hover{
  background-color:red;
}

/* Footer */
footer{
  background-color:#2c3e50;
  color:white;
  text-align:center;
  padding:15px;
  margin-top:30px;
}
```

Code Explanation

HTML5 is used to structure the content of a webpage using semantic elements such as <header>, <nav>, <section>, and <footer>. CSS3 is used to style and enhance the appearance of the webpage by applying colors, fonts, spacing, borders, shadows, gradients, layouts, and interactive hover effects.

In this experiment, a styled webpage titled "My Styled Webpage" is created. The webpage contains a header, navigation bar, three content cards for HTML5, CSS3, and JavaScript, and a footer.

CSS3 properties such as linear-gradient(), display: flex, box-shadow, border-radius, text-shadow, :hover, flex-wrap, and spacing properties are used to improve the design and user experience.

HTML5 Components Used

Component	Purpose
<header>	Defines the header section of the webpage
<nav>	Creates the navigation menu
<a>	Creates navigation links
<section>	Groups the main content
<div>	Creates individual content cards
<h1>, <h2>	Displays headings
<p>	Displays paragraphs
<button>	Creates interactive buttons
<footer>	Defines the footer section

CSS3 Properties Used

CSS Property	Purpose
background	Adds background styling
linear-gradient()	Creates a gradient background
color	Sets text color
font-family	Changes the font
text-align	Aligns text
padding	Adds space inside elements
margin	Adds space outside elements
display: flex	Creates a flexible layout
flex-wrap	Allows cards to move to the next line
gap	Creates space between cards
border	Adds borders
border-radius	Creates rounded corners
box-shadow	Adds shadow effects
text-shadow	Adds shadow to text
:hover	Applies styles when the user points to an element

TEST CASES AND EXECUTION DATA

Test Case	Component Tested	Action / Input	Expected Result	Actual Result	Status
TC-01	Header & Navigation	Open the webpage	Header and navigation links should be displayed correctly	Displayed correctly	Pass
TC-02	Content Cards	View HTML5, CSS3 and JavaScript cards	All three cards should be displayed with proper styling	Displayed correctly	Pass
TC-03	Hover Effects	Move pointer over navigation links and buttons	Hover styles should be applied	Hover effects worked correctly	Pass
TC-04	Responsive Layout	Resize the browser window	Cards should wrap according to available space	Cards wrapped correctly	Pass

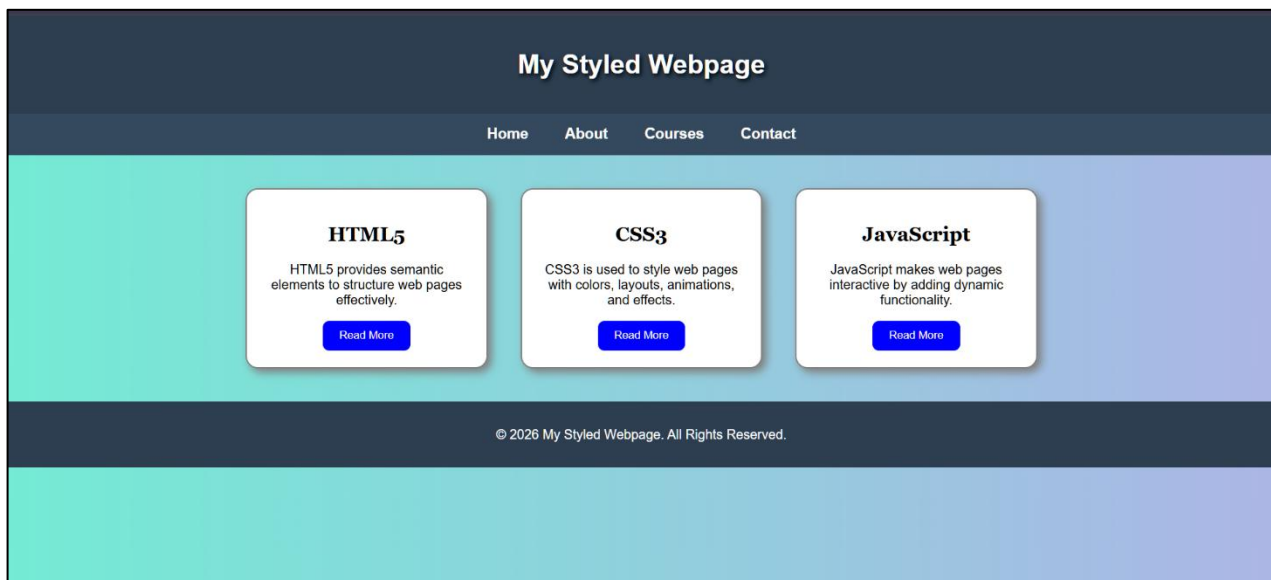
TC OUTPUT 01



TC OUTPUT 02 & 03 & 04:



Output:



RESULT

Thus, a styled webpage using HTML5 and CSS3 was successfully designed and developed. Semantic HTML5 elements and CSS3 properties such as gradients, Flexbox, borders, shadows, spacing, and hover effects were implemented successfully. The card-based layout was tested for responsiveness and the webpage was verified successfully.